



Vortex-induced vibration of bridge decks: Describing function-based model

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ABSTRACT

A describing function (DF)-based model is introduced for the simulation of vortex-induced vibration (VIV) of bridge decks. Similar to the linear frequency response function, the DF is the complex ratio of the first-order component of the nonlinear output to the harmonic input. The DF can be either identified using the forced vibration technique or based on the VIV nonlinear response time history. An iterative procedure is accordingly developed to predict the VIV response with the DF-based model, and an equivalent-damping-ratio-based simplified method is further proposed to efficiently obtain the limit cycle oscillation (LCO) amplitude of VIV. It is demonstrated that the conventional van der Pol-type model is equivalent to a special case of the DF-based model for VIV. Three case studies involving various cross-sections are utilized to validate the simulation accuracy and efficiency of the proposed DF-based model for typical features at VIV lock-in such as LCO and hysteresis phenomena. The vertical VIVs can be well captured by the DF-based model, while its capability of simulating the torsional VIVs requires further improvement. Furthermore, the predictive capability of DF-based model for vertical VIVs of bridge decks within a wide range of mass-damping conditions is highlighted.

1. Introduction

Vortex-induced vibration (VIV), as a typical flow-induced vibration of bluff bodies, may seriously impact the fatigue life of bridges and/or the traffic safety. VIV with relatively large response of limit cycle oscillation (LCO) occurs for a structure with low mass-damping parameter (i.e., the Scruton number, Sc) as the dominant vortex shedding frequency approaches its natural frequency. Both vertical and torsional VIVs have been observed in the wind tunnel tests of bridge decks (Ehsan and Scanlan, 1990; Diana et al., 2006; Marra et al., 2015; Xu et al., 2016a; Zhu et al., 2017) and full-scale measurements of several long-span flexible bridges (Frandsen, 2001; Fujino and Yoshida, 2002; Kim et al., 2013; Li et al., 2014a). Although a large number of studies have been focused on the VIV of circular cylinders, significant difference exists between the VIV features of circular cylinders and bluff bridge decks with long afterbody.

In the absence of an analytical solution of the flow around a bluff bridge deck, a closed-form description of the VIV phenomenon will remain mathematically intractable. A convenient approach for evaluating the VIV performance of a bridge is to conduct wind tunnel tests on a scaled model. However, these experimental techniques are limited to low

Reynolds numbers (Frandsen, 2001; Marra et al., 2017). Although the computational fluid dynamics (CFD)-based approaches have shown promising simulation capability of VIV, they have their own limitations resulting from a lack of robust turbulence models for engineering applications and expensive computational costs. As a result, the semi-empirical models have been advanced together with the experimentally identified parameters to assess the VIV of bridges.

Early work on engineering modeling of VIV for circular cylinders resulted in several widely-used semi-empirical models, e.g., the second-order nonlinear wake-oscillator (van der Pol-type oscillator) developed by Hartlen and Currie (1970) following the pioneering suggestion of Bishop and Hassan (1964a, b). To simplify the parameter identification procedure in the simulation of VIV for bridge decks, Scanlan (1981) suggested an explicit function of van der Pol-type oscillator for the vortex-induced forces on bridge decks. In Scanlan's VIV model, the LCO of VIV is essentially resulted from the prescribed aerodynamic damping terms. Following this idea, a number of studies have been conducted to further enhance the parameter identification procedure (Ehsan and Scanlan, 1990; Gupta et al., 1996; Marra et al., 2011) and/or to improve nonlinear aerodynamic damping expressions (Goswami et al., 1993a, 1993b; Diana et al., 2006; Zhu et al., 2013, 2017; Xu et al., 2018). It is

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well known that the VIV performance of a bridge deck is very sensitive to Sc , however, the van der Pol-type models seem to have rather limited predictive capability for VIV of bridge decks with different Sc (Ehsan and Scanlan, 1990; Marra et al. 2011, 2015). Actually, the parameters in Scanlan's model change significantly and irregularly with the mechanical damping ratio (and hence Sc).

Another approach to offer high fidelity simulations of bridge VIV at affordable computational demands is based on the reduced-order models (ROMs). Because there are much lower order degrees of freedom involved in a ROM as compared with CFD, it can be tailored to meet the demands placed by the fundamental physics of the bridge VIVs. Compared to the semi-empirical models, ROMs usually present better characterization and quantification of nonlinear bridge aerodynamics (Wu and Kareem, 2014, 2015). Wu and Kareem (2012, 2013) introduced a Volterra theory-based model to simulate the VIV of bridge decks, and the ROM methodology was later followed and improved by Xu et al. (2017). The basic premise of the Volterra theory-based model is that the nonlinear aerodynamics of VIV can be well captured by a sum of multi-dimensional convolution integrals of increasing order. Although the simulation results of bridge VIV based on the Volterra ROM is promising, the identification of higher-order kernels (model parameters) remains challenging. On the other hand, experimental and full-scale data of bridge VIV indicate that the deck response (vertical or torsional) at lock-in can be approximated as a mono-component signal. Therefore, the describing function (DF) method, with considerable advantage of being one dimensional in the frequency domain, is introduced here to more conveniently model the vertical and torsional VIVs of bridge decks. The DF method was originally developed in fields of nonlinear mechanics (Kryloff and Bogoliuboff, 1947) and nonlinear control theory (Gelb and Vander Velde, 1968). It has been widely used as a frequency-domain method for analyzing nonlinear aeroelastic systems (e.g., Mansour, 1972; Dowell and Ueda, 1984; Price et al., 1995). The DF-based model parameters used to relate the aerodynamic input and output components in the bridge VIV system, although amplitude dependent, is otherwise similar in form to the conventional linear frequency response function. Applications of the DF method to model the VIV of bridge decks in the present work actually indicate that the nonlinear LCO at VIV lock-in could be well represented by its first-order harmonic response. The results of three case studies involving various cross-sections demonstrate the high simulation accuracy and efficiency of the proposed DF-based model for typical features at VIV lock-in such as LCO and hysteresis phenomenon.

2. Governing equations of bridge deck VIV

The governing equations of the vertical and torsional VIVs of a spring-suspended bridge deck can be respectively expressed as

$$m(\ddot{h} + 2\xi_{h0}\omega_{h0}\dot{h} + \omega_{h0}^2 h) = F \quad (1a)$$

$$I(\ddot{\alpha} + 2\xi_{\alpha0}\omega_{\alpha0}\dot{\alpha} + \omega_{\alpha0}^2 \alpha) = M \quad (1b)$$

where m and I are the mass and moment of inertia per unit span length, respectively; h and α are the vertical and torsional displacements, respectively; over dot indicates the derivative with respect to time t ; ξ_{h0} and $\xi_{\alpha0}$ represent the vertical and torsional mechanical damping ratios, respectively; ω_{h0} and $\omega_{\alpha0}$ are the vertical and torsional natural circular frequencies, respectively; F and M are functions representing the vortex-induced lift force and torsional moment acting on the bridge deck per unit span length. F (or M) essentially consists of contributions induced by structural motion and vortex shedding, and it is conventionally expressed in the generalized van der Pol form to account for the nonlinear aerodynamic damping. For example, F in Scanlan's VIV model is explicitly expressed as a sum of the self-excited term and the forcing term (Scanlan, 1981)

$$F = \rho U^2 D \left\{ Y_1(K_h) \left[1 - \varepsilon(K_h) \frac{h^2}{D^2} \right] \frac{\dot{h}}{U} + Y_2(K_h) \frac{h}{D} + C_L(K_h) \sin(\omega_s t + \theta) \right\} \quad (2a)$$

where ρ = air density; U = mean wind speed; D = deck depth; $K_h = \omega_h B/U$ is the reduced frequency (where ω_h is the vertical circular vibration frequency); Y_1 , ε , Y_2 , and C_L are aerodynamic parameters to be identified; θ is the initial phase. Y_1 and ε are related to the linear and nonlinear components of the aerodynamic damping, respectively; Y_2 represents the aerodynamic stiffness term; C_L represents the forcing term at ω_s ($\omega_s = 2\pi S_t U/D$ is the Strouhal circular frequency, where S_t = Strouhal number). Experimental observations show that, in the lock-in region, the forcing term is negligible compared to the motion-induced force (Ehsan and Scanlan, 1990). Since the major purpose for the VIV analysis of bridge decks is to predict the large amplitude oscillations at lock-in, the forcing term may be dropped. As a result, Eq. (2a) can be simplified as

$$F = \rho U^2 D \left\{ Y_1(K_h) \left[1 - \varepsilon(K_h) \frac{h^2}{D^2} \right] \frac{\dot{h}}{U} + Y_2(K_h) \frac{h}{D} \right\} \quad (2b)$$

In addition, experimentally identified results indicate that there is only insignificant variation between ω_{h0} and ω_h at lock-in (Ehsan and Scanlan, 1990; Marra et al., 2011, 2015). Therefore, the aerodynamic stiffness term can also be dropped. Experimental observations show that the vertical (or torsional) VIV response in the lock-in region can be approximated as a mono-component signal at a frequency [denoted as ω_h (or ω_α) for vertical (or torsional) case] very close to ω_{h0} (or $\omega_{\alpha0}$). Accordingly, the VIV system is usually assumed to be a weakly nonlinear system which is dominated by the first-order harmonic in the vortex-induced force. This approximation also indicates that only the first-order harmonic of the VIV response is taken as an input to the VIV system. The nonlinear relation between the uni-dimensional input and output components at the same frequency can be well characterized using the amplitude-dependent DF-based model.

3 DF-based mathematical model for VIV of bridge decks

The DF method arose naturally from the equivalent linearization theory in solving nonlinear ordinary differential equations (Kryloff and Bogoliuboff, 1947) and was developed to a high degree of sophistication in the nonlinear control theory (Gelb and Vander Velde, 1968). It is an approximation technique for analyzing nonlinear systems, in which the nonlinearity is simply represented by transfer functions (from input to output) with amplitude-dependent features. Although DF method is applicable for complicated input types, the most widely discussed and used DF is the sinusoidal input DF (SIDF), which is defined as the complex ratio of the first-order harmonic in the output to the sinusoidal input of the system. The SIDF method is suitable for analyzing nonlinear systems which can be represented by their quasi-linear approximations (neglecting the higher-order components in the input and output) with sufficient accuracy.

Consider a nonlinear time-invariant system excited by a sinusoidal input $x = a \sin(\omega t + \theta)$, the output may be approximated as

$$y = \sum_{n=1}^p y_n = \sum_{n=1}^p A_n \sin[n(\omega t + \theta) + \beta_n] \quad (3)$$

where p is the highest harmonic order in the output; y_n ($n = 1 \sim p$) is the n th-order harmonic of y ; A_n is the amplitude of y_n ; $[n(\omega t + \theta) + \beta_n]$ is the phase of y_n . For a linear system, A_1/a and β_1 are functions of ω . However, investigations on a number of nonlinear systems including the aeroelastic system of bridge decks (Scanlan, 1997; Noda et al., 2003; Diana et al., 2006; Xu et al., 2016b) show that A_n/a and β_n are also dependent on the input amplitude a .

For a wide class of weakly nonlinear systems, Eq. (3) is dominated by the first-order harmonic. Similar with the frequency response function

widely used for linear systems, a transfer function (i.e., SIFD) can be defined as the complex ratio of the first-order harmonic in Eq. (3) to x , i.e.,

$$N(a, \omega) = \frac{A_1 e^{j(\omega t + \theta + \beta_1)}}{a e^{j(\omega t + \theta)}} = \frac{A_1}{a} e^{j\beta_1} = N_r(a, \omega) + jN_i(a, \omega) \quad (4)$$

where $j = \sqrt{-1}$; N_r and N_i are the real and image parts of the SIFD, which

$$m(\ddot{h} + 2\xi_{h0}\omega_{h0}\dot{h} + \omega_{h0}^2 h) = \rho U^2 D \left\{ \frac{Y_2(K_h)}{D} h + \left[Y_1(K_h) \frac{\omega_h}{U} \left(1 - \frac{\varepsilon(K_h)(A_h/D)^2}{4} \right) \right] \frac{\dot{h}}{\omega_h} \right\} \quad (9)$$

can be respectively calculated as

$$N_r(a, \omega) = \frac{1}{a\pi} \int_0^{2\pi} y \sin(\omega t + \theta) d(\omega t) \quad (5a)$$

$$N_i(a, \omega) = \frac{1}{a\pi} \int_0^{2\pi} y \cos(\omega t + \theta) d(\omega t) \quad (5b)$$

The SIFD $N(a, \omega)$ can be viewed as a simple extension of the frequency response function for a linear system with amplitude-dependent feature. The output of the nonlinear system to any sinusoidal input x can be approximated using the SIFD as

$$y = N_r(a, \omega)x + N_i(a, \omega)\frac{\dot{x}}{\omega} \quad (6)$$

In the bridge aeroelastic simulation, it is obvious that the SIFD input and output are the deck motion and wind-induced force, respectively. More specifically, the vertical or torsional motion input in the VIV lock-in region can be approximated as a mono-component signal at ω_h or ω_a . Accordingly, the SIFD-based F and M in the lock-in region may be respectively approximated as

$$F = \rho U^2 D \left[N_r^h(A_h/D, \omega_h) h + N_i^h(A_h/D, \omega_h) \frac{\dot{h}}{\omega_h} \right] \quad (7a)$$

$$M = \rho U^2 D^2 \left[N_r^a(A_a/D, \omega_a) \alpha + N_i^a(A_a/D, \omega_a) \frac{\dot{\alpha}}{\omega_a} \right] \quad (7b)$$

where N_r^h (or N_r^a) and N_i^h (or N_i^a) are SIFD parameters related to the real and image parts of the vertical (or torsional) force output, respectively. Hence, the equations of motion for the vertical and torsional VIVs of a bridge deck can be respectively expressed as

$$m(\ddot{h} + 2\xi_{h0}\omega_{h0}\dot{h} + \omega_{h0}^2 h) = \rho U^2 D \left[N_r^h(A_h/D, \omega_h) h + N_i^h(A_h/D, \omega_h) \frac{\dot{h}}{\omega_h} \right] \quad (8a)$$

$$I(\ddot{\alpha} + 2\xi_{a0}\omega_{a0}\dot{\alpha} + \omega_{a0}^2 \alpha) = \rho U^2 D^2 \left[N_r^a(A_a/D, \omega_a) \alpha + N_i^a(A_a/D, \omega_a) \frac{\dot{\alpha}}{\omega_a} \right] \quad (8b)$$

The unknown SIFD parameters N_r^h and N_i^h (or N_r^a and N_i^a) can be straightforwardly obtained by using curve fitting or Fourier expansion of the time history of F (or M) at lock-in, where only the first-order harmonic is concerned. It is noted that the dimensionless form of Eq. (7) relates the SIFD parameters for a specific vibration amplitude to the conventional aerodynamic transfer functions for bridge aeroelasticity. Therefore, the well-established identification techniques for aerodynamic transfer functions based on wind tunnel tests or CFD simulation can be utilized to facilitate the use of the SIFD-based VIV model. More

detailed identification procedure of the SIFD parameters at VIV lock-in will be introduced in the following section.

It is worth mentioning that the van der Pol-type model (in the lock-in region) can be transformed into the form of Eq. (8a) or (8b) through an equivalent linearization (Bogoliubov and Mitroposki, 1961). For example, Scanlan's VIV model [Eq. (2)] can be transformed into the form of Eq. (8a) as

Comparing Eq. (8a) with Eq. (9), it is noted that Scanlan's VIV model is equivalent to a special case of the SIFD-based VIV model, in which N_r^h is quadratically related and to A_h/D , while N_i^h is a constant for each dimensionless oscillation frequency (reduced frequency). For a more complex bridge aeroelastic system involving dual harmonic or random motion inputs, a comprehensive DF-based model can be deduced according to their relations with the frequency-domain Volterra transfer function of the system (Peyton-Jones and Billings, 1991; Carassale et al., 2014; Wu and Kareem, 2014, 2015).

4 Extraction of SIFD parameters

As briefly mentioned in the preceding section, the SIFD parameters can be straightforwardly identified using the first-order harmonic of F (or M) at lock-in, which is obtained through curving fitting of the output data of the forced vibration tests or numerical simulations (with sinusoidal motions). In the case that a closed-form description of the VIV phenomenon exists, Fourier expansion of F (or M) at lock-in can be also employed. The forced vibration-based identification approach is very computational expensive and time consuming since for each specific reduced frequency, experimental tests (or numerical simulations) with a number of different vibration amplitudes are needed. To improve the identification efficiency of SIFD parameters, a convenient procedure based on the experimentally recorded or numerically simulated VIV decay/growth-to-resonance signal is developed in this section. For sake of brevity, only the identification process for the SIFD parameters related to the vertical VIV response (N_r^h and N_i^h) is presented here.

To identify the SIFD parameters N_r^h and N_i^h , the amplitude-dependent circular frequency ω_h (A_h/D) and damping ratio ξ_h (A_h/D) of the VIV decay/growth-to-resonance signal $h(t)$ need to be first extracted using the method developed in Zhang and Xu (2018), which is briefly summarized here for completeness:

- (i) Extract $A_h(t)$ from $h(t)$ through cubic spline fittings, and calculate the instantaneous circular frequency $\omega_h(t)$ by using the normalized Hilbert transform (NHT) method (Huang et al., 2009).
- (ii) Calculate ω_h (A_h/D) through a least-square polynomial fitting on the obtained $[A_h(t)/D, \omega_h(t)]$, i.e.,

$$[A_h(t)/D, \omega_h(t)]^{\text{polynomial}} \rightarrow \text{fitting} \omega_h(A_h/D) \quad (10a)$$

- (iii) Calculate the instantaneous damping ratio $\xi_h(t)$ of $h(t)$ through

$$\xi_h(t) = \frac{\ln[A_h(t_n)/A_h(t_{n-1})]}{2\pi} \quad (10b)$$

where t_n and t_{n-1} represent two successive sampling time instants. Although Eq. (10b) is an approximation of $\xi_h(t)$, it has been demonstrated

that the relative error is lower than 0.5% for $\xi_h(t) < 10\%$.

- (iv) Calculate $\xi_h(A_h/D)$ through a least-square polynomial fitting on the obtained $[A_h(t)/D, \xi_h(t)]$, i.e.,

$$[A_h(t)/D, \xi_h(t)]^{\text{polynomial}} \rightarrow \xi_h(A_h/D) \quad (10c)$$

According to Eq. (8a), the analytical solutions of $\omega_h(A_h/D)$ and $\xi_h(A_h/D)$ can be respectively expressed as

$$\omega_h(A_h/D) = \sqrt{\omega_{h0}^2 - \frac{\rho U^2 D N_i^h}{m}} \quad (11a)$$

$$\xi_h(A_h/D) = \xi_{h0} - \frac{\rho U^2 D N_i^h}{2m\omega_{h0}\omega_h} \quad (11b)$$

Consequently, N_r^h and N_i^h can be respectively calculated as

$$N_r^h(A_h/D) = \frac{m[\omega_{h0}^2 - \omega_h^2(A_h/D)]}{\rho U^2 D} \quad (12a)$$

$$N_i^h(A_h/D) = \frac{2m\omega_{h0}\omega_h[\xi_{h0} - \xi_h(A_h/D)]}{\rho U^2 D} \quad (12b)$$

N_r^h and N_i^h can be obtained using Eqs. (12a) and (12b), respectively.

5 Prediction of VIV response in lock-in region

An iteration prediction procedure for the VIV response based on the SIFD parameters (related to both aerodynamic damping and stiffness) is developed in this section. In addition, a simplified prediction procedure for the VIV amplitude based on the aerodynamic damping SIFD parameter is also introduced. Only the prediction of vertical VIV is presented here for sake of brevity.

5.1. Predicting VIV response using N_r^h and N_i^h

For a specific bridge deck, m , ξ_{h0} , ω_{h0} , B , and D can be predetermined. The vertical vortex-induced force is a function of N_r^h and N_i^h as shown in Eq. (7a), and ω_h is an unknown value very close to ω_{h0} . Once N_r^h and N_i^h are identified for a sufficiently wide range of reduced frequencies, the vertical VIV response at each U can be calculated through an iterative procedure as follows:

Step 1: Set ω_{h0} as the initial value of ω_h . Substitute $N_r^h(A_h/D, \omega_h)$ and $N_i^h(A_h/D, \omega_h)$ into the vertical equation of motion, i.e., Eq. (8a). Solve Eq. (8a) (from a selected initial condition) by using the Newmark- β method, and calculate the resulting circular frequency $\omega_{h,1}$. The difference between $\omega_{h,1}$ and ω_{h0} is denoted as e_1 , i.e., $e_1 = \omega_{h,1} - \omega_{h0}$.

Step j ($j \geq 2$): Set $\omega_h = \omega_{h,j-1}$. Substitute $N_r^h(A_h/D, \omega_h)$ and $N_i^h(A_h/D, \omega_h)$ into Eq. (8a). Repeat the aforementioned procedure, and calculate the resulting circular frequency $\omega_{h,j}$. The difference between $\omega_{h,j}$ and $\omega_{h,j-1}$ is denoted as e_j , i.e., $e_j = \omega_{h,j} - \omega_{h,j-1}$.

The iteration terminates at the m th step if $e_m \leq \varepsilon$, where ε is a pre-specified small number (e.g., $\varepsilon = 0.01$). As ω_h is very close to ω_{h0} , the iterative process converges rapidly.

5.2. Predicting VIV amplitude using N_i^h

As mentioned in Section 2, the difference between the lock-in vibration frequency and the natural frequency of the structure is insignificant. Therefore, the aerodynamic stiffness term can be neglected and Eq. (8a) is accordingly simplified as

$$m(\ddot{h} + 2\xi_{h0}\omega_{h0}\dot{h} + \omega_{h0}^2 h) = \rho U^2 D N_i^h(A_h/D, \omega_h) \frac{\dot{h}}{\omega_h} \quad (13)$$

The equivalent damping ratio of the system governed by Eq. (13) is dependent on A_h/D and can be expressed as

$$\xi_h(A_h/D) = \xi_{h0} + \xi_{h,aero} = \xi_{h0} - \frac{\rho U^2 D N_i^h(A_h/D, \omega_h)}{2m\omega_{h0}\omega_h} \quad (14)$$

where $\xi_{h,aero}$ is the equivalent aerodynamic damping ratio.

It is noted that the vertical VIV amplitude at a specific U corresponds to the intersection point between the $\xi_{h,aero}$ versus A_h/D curve and the horizontal line of $\xi = \xi_{h0}$ with $R < 0$, where R is the slope of the $\xi_{h,aero}$ versus A_h/D curve. As illustrated in Fig. 1, for an intersection point with $R < 0$ (e.g., P_2), $\xi_{h0} + \xi_{h,aero} < 0$ for $A_h < A_{h2}$ while $\xi_{h0} + \xi_{h,aero} > 0$ for $A_h > A_{h2}$, and hence the amplitude tends to A_{h2} regardless of the initial amplitude (within a limited range around A_{h2}). However, for an intersection point with $R > 0$ (e.g., P_1), $\xi_{h0} + \xi_{h,aero} > 0$ for $A_h < A_{h1}$ while $\xi_{h0} + \xi_{h,aero} < 0$ for $A_h > A_{h1}$, and hence the amplitude always decays or diverges away from A_{h1} regardless of the initial amplitude (within a limited range around A_{h1}). Consequently, P_2 is a stable state and accordingly A_{h2} is the VIV amplitude, whereas P_1 is an unstable state. Since the equivalent damping ratio around the P_1 state is very close to zero, the vibration decays or diverges very slowly. Actually, the vibration may occasionally stay around the P_1 state (i.e., $A_h \approx A_{h1}$) for quite a long duration before finally goes to the other state (e.g., P_2), as reported in Goswami et al. (1993a) for the wind tunnel tests of a circular cylinder VIV.

6. Case studies

To demonstrate the simulation accuracy and efficiency of the SIFD-based model for typical features at VIV lock-in such as LCO and hysteresis phenomenon, three case studies involving rectangular and real bridge deck sections are investigated in this section.

6.1. VIV of a $B/D = 4$ rectangular cylinder

6.1.1. Vertical LCO at VIV lock-in

The vertical VIV responses of a rectangular cylinder (with $B = 0.30$ m, $D = 0.075$ m, $m = 7.3$ kg/m, and $\omega_{h0} = 49.5$ rad/s) from Marra et al. (2015) are utilized to extract the SIFD parameters. The vertical mechanical damping ratio ξ_{h0} and natural circular frequency ω_{h0} were identified from free decay vibration tests in still air at 90° angle of attack (to reduce the interaction between the cylinder and its surrounding air) with vibration amplitudes lower than 0.5 mm. The experimentally recorded vertical VIV responses for $K_{h0} = 3.086$, $Sc = 78.1$ and for $K_{h0} = 2.945$, $Sc = 21.7$ are shown in Fig. 2(a) and (b), respectively, and the corresponding $\omega_h(A_h/D)$ and $\xi_h(A_h/D)$ are shown in Fig. 3(a) and Fig. 4(a), respectively. It is noted that ω_h is very close to ω_{h0} , indicating

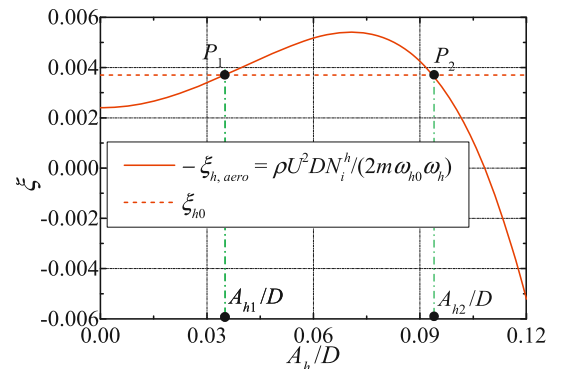


Fig. 1. Aerodynamic and mechanical damping ratios for a VIV system.

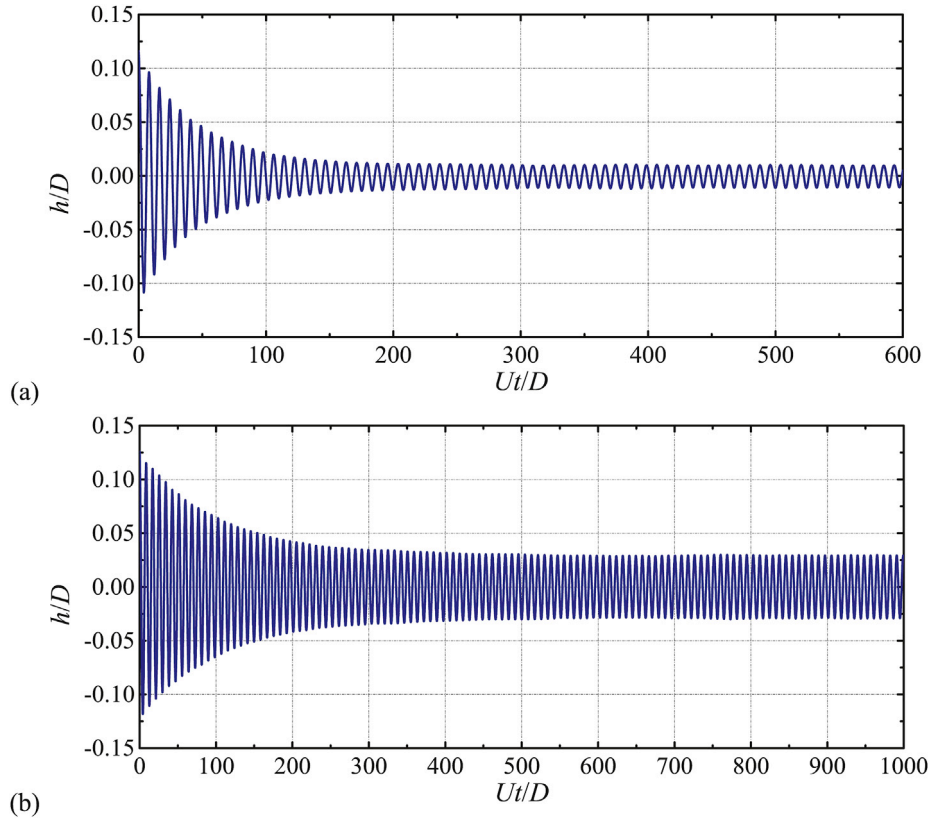


Fig. 2. Dimensionless vertical displacement time histories of a $B/D = 4$ rectangular section at VIV lock-in (data from Marra et al., 2015): (a) $K_{h0} = 3.086$, $Sc = 78.1$; (b) $K_{h0} = 2.945$, $Sc = 21.7$.

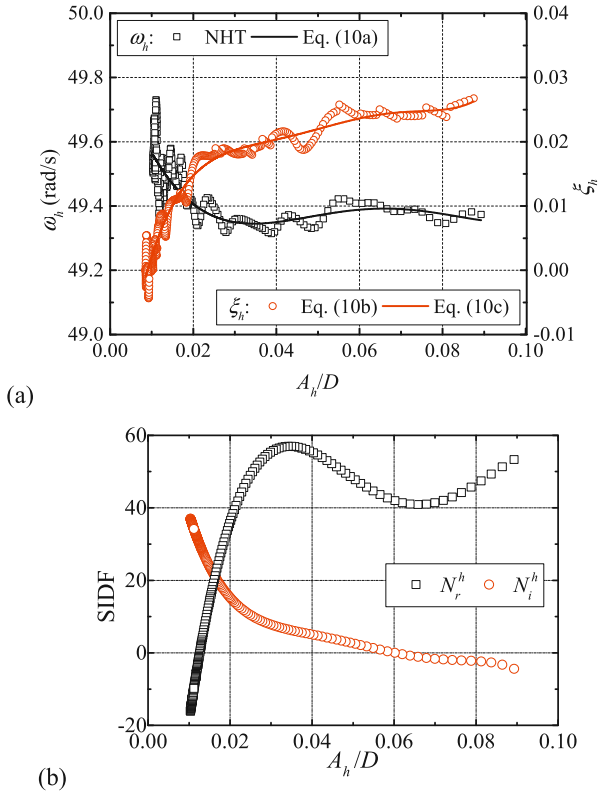


Fig. 3. Identified parameters of VIV system (a $B/D = 4$ rectangular section at $K_{h0} = 3.086$, $Sc = 78.1$): (a) ω_h (A_h/D) and ξ_h (A_h/D); (b) N_r^h and N_i^h .

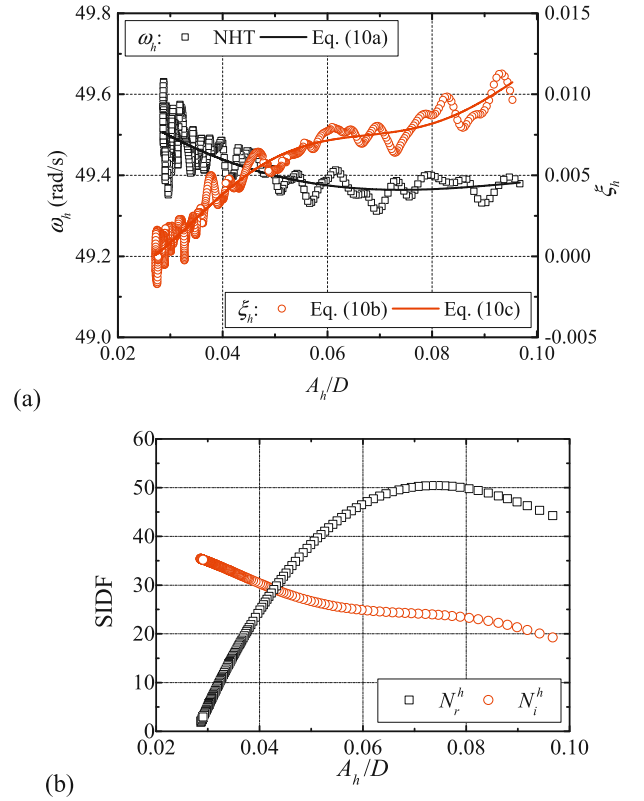


Fig. 4. Identified parameters for VIV system (a $B/D = 4$ rectangular section at $K_{h0} = 2.945$, $Sc = 21.7$): (a) ω_h (A_h/D) and ξ_h (A_h/D); (b) N_r^h and N_i^h .

that the aerodynamic stiffness effects on the VIV response are insignificant. On the other hand, ξ_h vary significantly with A_h/D , indicating that the aerodynamic damping exhibits obvious nonlinear features. The observable fluctuations in the NHT-based ω_h (A_h/D) and Eq. (10b)-based ξ_h (A_h/D) are mainly ascribed to the higher-order harmonics in the VIV responses, noises in the recorded signals, and cubic spline fitting errors.

N_r^h and N_i^h for $K_{h0} = 3.086$, $Sc = 78.1$ and for $K_{h0} = 2.945$, $Sc = 21.7$ are calculated according to Eqs. (12a) and (12b), respectively, and the results are shown in Figs. 3(b) and Fig. 4(b), respectively. Although N_r^h varies remarkably with A_h/D , its effects on the VIV response is very small due to the insignificant difference between ω_h (A_h/D) and ω_{h0} . To verify the identification accuracy of N_r^h and N_i^h , the obtained results are substituted into Eq. (8a) to calculate the VIV responses with the Newmark- β method using the first sampled peaks as initial conditions. The calculated responses are presented together with the experimental records in Fig. 5. It is shown that the calculated time histories agree very well with the recorded ones, indicating a high identification accuracy of N_r^h and N_i^h . In addition, the calculated VIV responses with $N_r^h = 0$ are also shown in Fig. 5. The comparison of results with and without consideration of N_r^h demonstrates the aerodynamic stiffness effects on VIV response are insignificant.

The VIV amplitudes with $K_{h0} = 3.086$ and $K_{h0} = 2.945$ for a wide range of Sc are predicted using the SIFD-based model only involving N_i^h , and based on Scanlan's VIV model involving model parameters ($Y_1 = 54.2$, $Y_2 = 0$, $\varepsilon = 10800$ for $K_{h0} = 3.086$ and $Y_1 = 17.2$, $Y_2 = 0$, $\varepsilon = 1942$ for $K_{h0} = 2.945$) experimentally extracted through the decay-to-resonance method (Ehsan and Scanlan, 1990). The simulation results are compared to the experimental records, as shown in Fig. 6. It can be seen that the SIFD-based results have been significantly improved compared

to those of Scanlan's model. In Fig. 6(b), the results of the SIFD-based model are not available for $K_{h0} = 2.945$, $Sc > 21.7$ due to a lack of the SIFD parameters for $A_h/D < 0.03$ (The lowest dimensionless amplitude of the corresponding decay-to-resonance signal is around 0.03). The growth-to-resonance test at $K_{h0} = 2.945$, $Sc = 21.7$ is necessary to obtain the SIFD parameters for $A_h/D < 0.03$. It is noted that this shortcoming can be conveniently overcome by a decay-to-resonance test at the highest Sc or a grow-to-resonance test at the lowest Sc . Due to the prescribed form of the aerodynamic damping, Scanlan's VIV model presents lower simulation accuracy compared to the SIFD-based model in predicting the LCO amplitudes at various mass-damping conditions. The relatively lower simulation accuracy of the SIFD-based model for $K_{h0} = 3.086$, $Sc < 10$ mainly results from the identification errors of N_i^h , where the cubic spline fitted instantaneous amplitude may be not accurate enough to extract the instantaneous damping ratio for the rapidly decaying response of $K_{h0} = 3.086$ and $Sc = 78.1$ (note that both high Sc and aerodynamic damping ratio may result in rapidly decaying of vibration signal, and hence reduce the accuracy of parameter identification with related signal segment). In addition, the amplitude-dependent characteristics of the mechanical damping and still-air aerodynamic damping may further reduce the identification accuracy of N_i^h .

6.1.2. Torsional LCO at VIV lock-in

Scanlan (1981) indicated that the same model (e.g., the van der Pol-type model) can be used to simulate both vertical and torsional VIV responses of bridge decks. The torsional VIV of a $B/D = 4$ rectangular cylinder is investigated here to demonstrate the ability of the proposed SIFD-based method in simulating the torsional VIV response. Experimental measurements or numerical simulations on the torsional VIVs of $B/D = 4$ rectangular cylinders are available in Matsumoto et al. (2008)

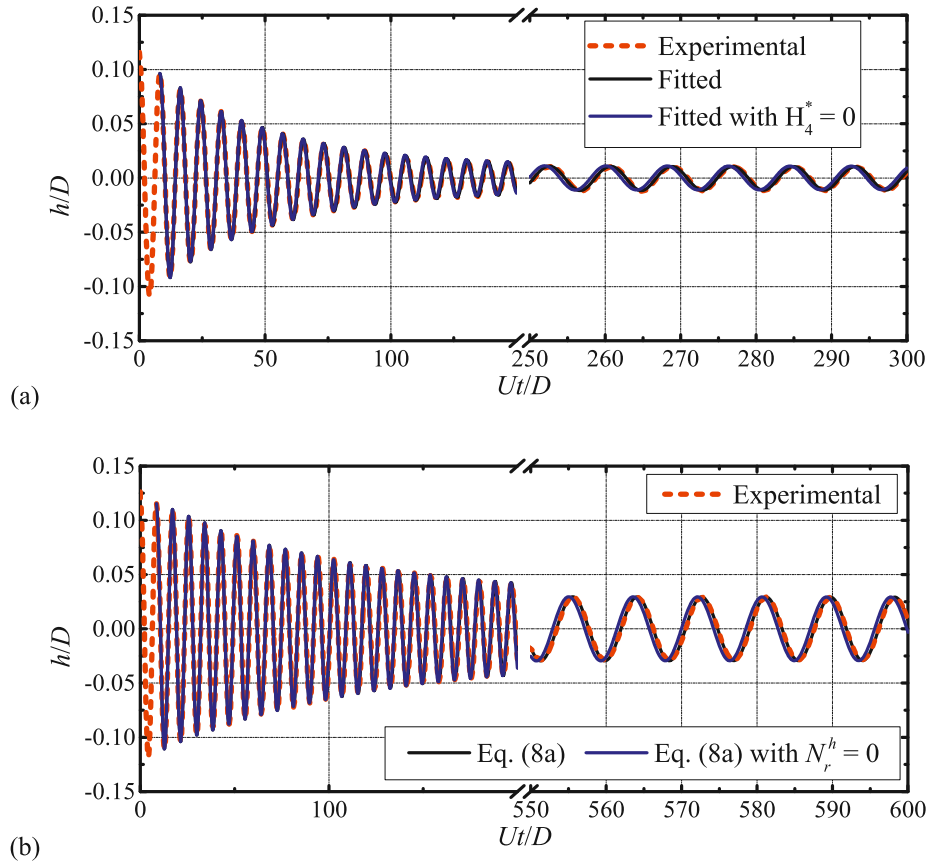


Fig. 5. Comparisons between experimental and simulated vertical displacement time histories of a $B/D = 4$ rectangular section at VIV lock-in: (a) $K_{h0} = 3.086$, $Sc = 78.1$; (b) $K_{h0} = 2.945$, $Sc = 21.7$.

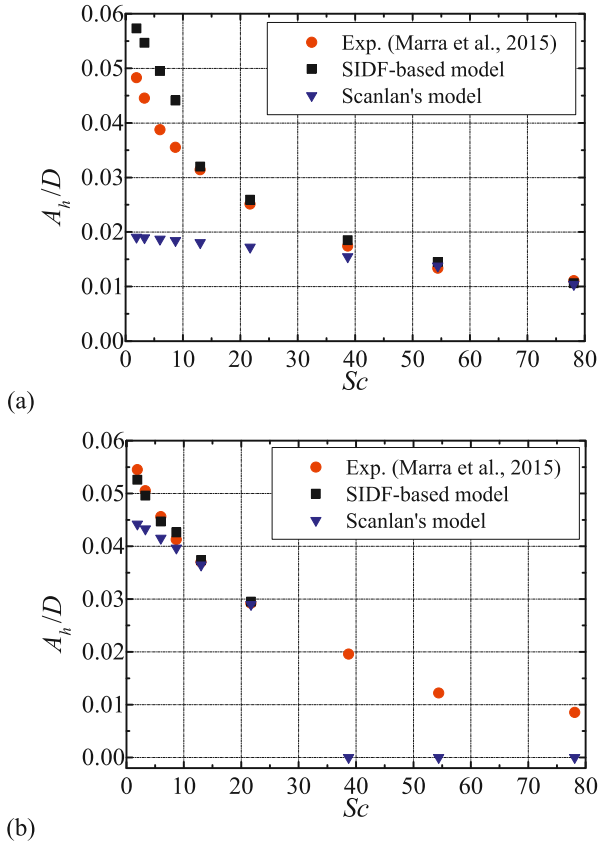


Fig. 6. Comparisons between experimental and simulated vertical VIV amplitudes of a $B/D = 4$ rectangular section: (a) $K_{h0} = 3.086$; (b) $K_{h0} = 2.945$.

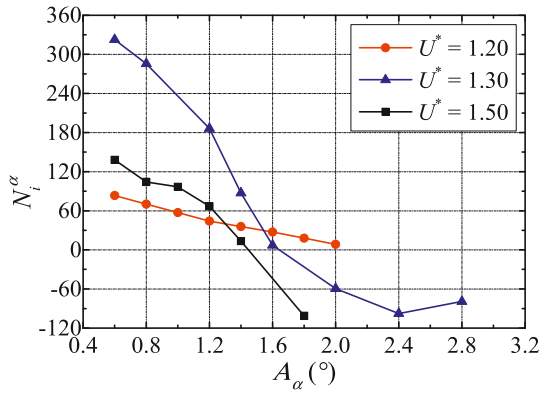


Fig. 7. N_i^a of a $B/D = 4$ rectangular section at different U^* .

$[Sc = 4\pi I_{\xi a 0}/(\rho B^2 D^2) = 0.49]$ and Shimada and Ishihara (2012) ($Sc = 5.30$).

The SIFD parameters N_r^h and N_i^h of the rectangular section at reduced wind speeds $U^* = 2\pi U/\omega_{a0}B = 1.20, 1.30$, and 1.50 (where $\omega_{a0} = 4\pi$ rad/s and $B = 0.12$ m) are extracted based on the numerically forced vibration simulations with a time step of 2×10^{-4} s (Xu et al., 2014). As shown in Fig. 7, N_i^a varies remarkably with the vibration amplitude. The results of N_r^a are not shown because the aerodynamic stiffness effects are insignificant. The torsional LCO amplitudes at VIV lock-in at $U^* = 1.20, 1.30$, and 1.50 for $Sc = 0.49$ and 5.30 are simulated based on the equivalent damping ratio resulting from N_i^a (Note that N_i^a are extracted based on numerically forced vibration simulations). The results of SIFD-based model are compared with those of two-dimensional (2D) $k - \epsilon$ numerical simulations in Shimada and Ishihara (2012), as shown in Fig. 8,

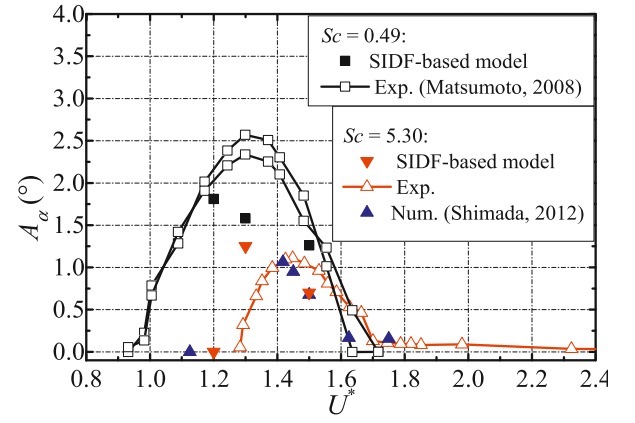


Fig. 8. Comparison between experimental, numerical, and simulated torsional VIV amplitudes of a $B/D = 4$ rectangular section.

indicating that the proposed scheme is capable of predicting the torsional VIV response of the rectangular section. However, there is relatively large discrepancy between the experimental measurements (Matsumoto et al., 2008) and SIFD-based simulations. The differences between the experimental and SIFD-based results may be ascribed to the errors commonly existing in the wind tunnel tests (e.g., fabrication errors of model and measurement errors of mechanical damping ratio) and numerically forced vibration simulations (Ying et al., 2017). Further researches are required to explain the discrepancy and improve the simulation accuracy for torsional VIV.

6.2. Vertical VIV of a bridge deck

The second example utilizes the vertical VIV response of a bridge deck section model ($B = 0.467$ m, $D = 0.055$ m, $m = 11.361$ kg/m, $\omega_{h0} = 26.3$ rad/s) described in Xu et al. (2018) to demonstrate the ability of the SIFD-based model in simulating the VIV of real bridge decks. The experimentally measured vertical VIV amplitudes in the lock-in region for $Sc = 24.53$ and 55.19 are shown in Fig. 9, where $U^* = 2\pi U/\omega_{h0}B$ is the reduced wind speed. It is noted that the damping ratio of the bridge deck in still air condition varies slightly with the vibration amplitude (Xu et al., 2018), and the mean value of the still-air damping ratio is used to calculate the Sc . The vertical VIV responses at $U^* = 2.34$ for $Sc = 24.53$ and 55.19 are presented in Fig. 10, and the identified N_r^h and N_i^h are given in Fig. 11.

The results extracted with two different Sc agree well with each other, indicating that the SIFD are independent of the mass-damping condition of the section model. N_r^h and N_i^h for $Sc = 55.19$ are substituted into Eq. (8a), and the corresponding VIV response is calculated based on the

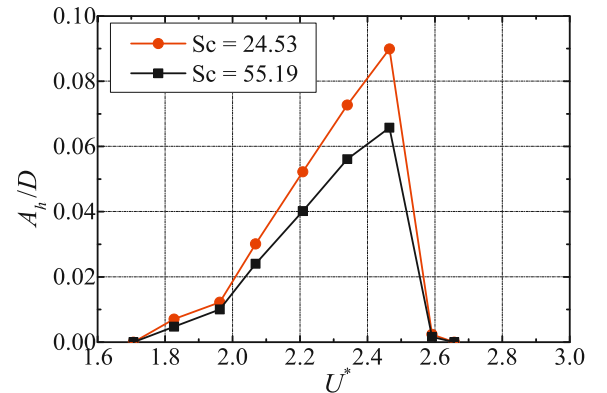


Fig. 9. Vertical VIV amplitudes of a bridge deck at different Sc (data from Xu et al., 2018).

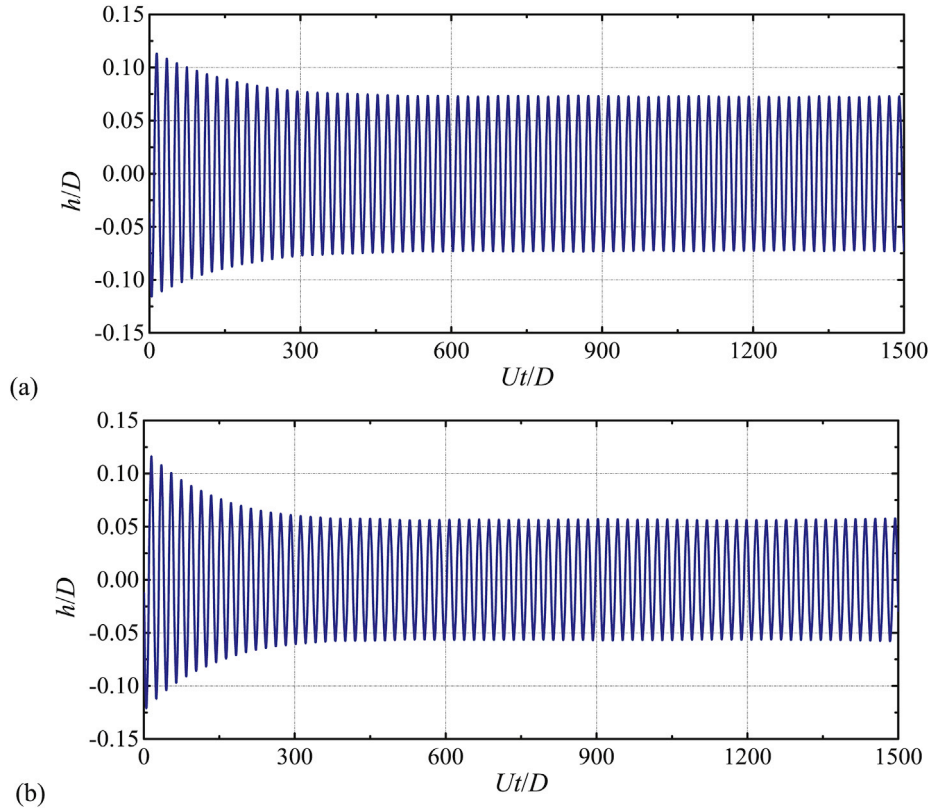


Fig. 10. Dimensionless vertical displacement time histories of a bridge deck at VIV lock-in ($U^* = 2.34$, data from Xu et al., 2018): (a) $Sc = 24.53$; (b) $Sc = 55.19$.

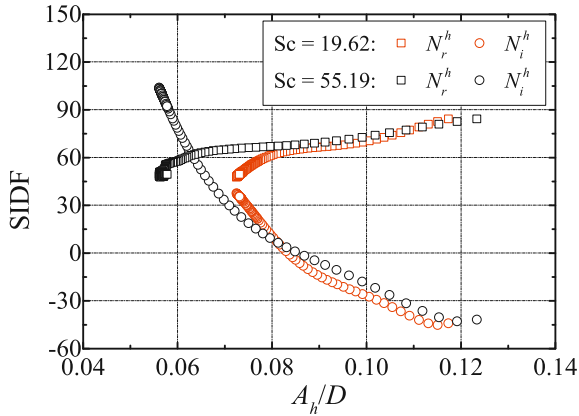


Fig. 11. N_r^h and N_i^h for a bridge deck at $U^* = 2.34$.

Newmark- β method using the first sampled peaks as initial condition. As shown in Fig. 12, the simulated time history agrees very well with the experimental record. The VIV response without consideration of N_r^h , as presented in Fig. 12, further demonstrates that the effects of the aerodynamic stiffness term are insignificant.

N_i^h at various reduced wind speeds ($U^* = 2.07, 2.21, 2.34$, and 2.47) are extracted from the corresponding decay-to-resonance test at $Sc = 55.19$, and the results are shown in Fig. 13. It is noted that N_i^h exhibit significant nonlinear feature, and vary in a similar trend with A_h/D . The LCO amplitudes of the bridge deck at VIV lock-in for $Sc = 24.53$ and 55.19 are calculated based on N_i^h (extracted from the decay-to-resonance test at $Sc = 55.19$), and the simulation results are compared to the experimental records as shown in Fig. 14. It can be seen that the SIFD-based simulations agree satisfactorily with the experimental results for both Sc .

6.3. Hysteresis behavior of bridge deck at VIV lock-in

An important phenomenon that may exist for VIV systems is the hysteresis behavior, which means that, in a certain wind speed range, two stable states are possible (e.g., a static state or a state of LCO, depending on the initial condition) for a specific wind speed. The hysteresis behavior in VIV systems has been investigated since Feng's tests on circular and D-section cylinders (Feng, 1968), and it has been recently observed in the wind tunnel tests of bridge VIVs (Li et al., 2014b). Although Currie, 1972 showed that the hysteresis behavior of a VIV system can be reproduced by introducing a small nonlinear mechanical stiffness term, a widely-used method to simulate the hysteresis behavior is to modify the original wake oscillator model by considering higher-order (e.g., 5th- or 7th-order) nonlinear aerodynamic damping terms (Landl, 1975; Wood and Parkinson, 1977; Facchinetti et al., 2004). The similar idea can be applied to enable the van der Pol-type model to simulate the hysteresis phenomenon. For example, by introducing a 5th-order nonlinear aerodynamic damping term (Landl, 1975; Wood and Parkinson, 1977) and neglecting the aerodynamic stiffness term in Scanlan's VIV model, a phenomenological model for vertical VIV response can be expressed as

$$m(\ddot{h} + 2\xi_{h0}\omega_{h0}\dot{h} + \omega_{h0}^2 h) = \rho U^2 D \left[Y_1(K_h) \left(1 - \varepsilon_1(K_h) \frac{h^2}{D^2} - \varepsilon_2(K_h) \frac{h^4}{D^4} \right) \frac{\dot{h}}{U} \right] \quad (15)$$

To investigate the ability of the proposed SIFD-based VIV model in capturing the hysteresis phenomenon, a VIV system represented by Eq. (15) with parameters listed in Table 1 is utilized as a reference system. m , ξ_0 , B , D , and K in Table 1 corresponds to a bridge deck section model in Scanlan (1998), while Y_1 , ε_1 , and ε_2 are elaborately selected to produce the hysteresis phenomenon. It should be stated that the right-hand side of Eq. (15) with the aerodynamic parameters in Table 1 does not necessarily represent the aerodynamic effects of a real bridge deck, however, it is

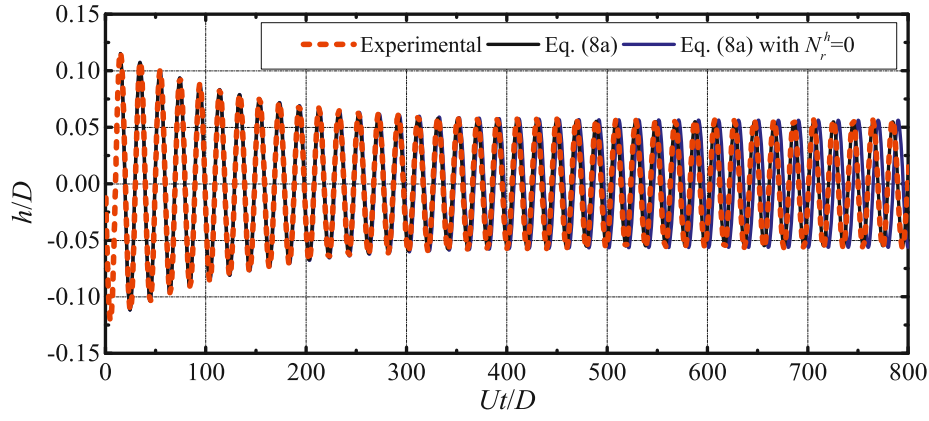


Fig. 12. Comparison between experimental and simulated vertical displacement time histories of a bridge deck at VIV lock-in ($U^* = 2.34$, $Sc = 55.19$).

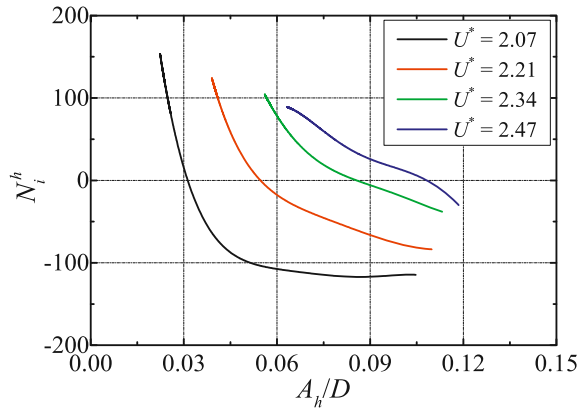


Fig. 13. N_i^h of a bridge deck at different U^* .

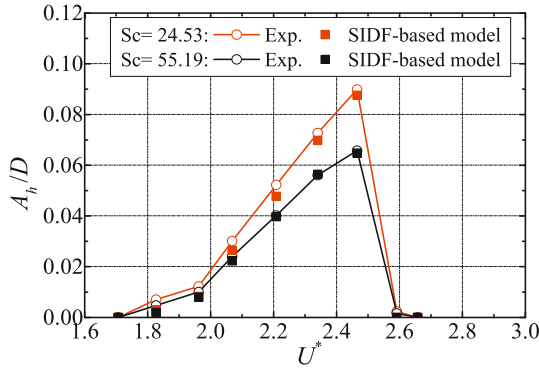


Fig. 14. Comparison between experimental and simulated vertical VIV amplitudes of a bridge deck.

utilized herein as it serves as a convenient phenomenological model with hysteresis behavior. Eq. (15) is numerically solved by using the 4th-order Runge-Kutta method with a dimensionless time step of 0.01 and initial conditions of $[h = A_0, \dot{h} = 0]$, where A_0 is increased from $0.005D$ to $0.150D$ with a $0.001D$ incremental. The vertical VIV amplitudes for various initial conditions are shown in Fig. 15, and the vertical VIV

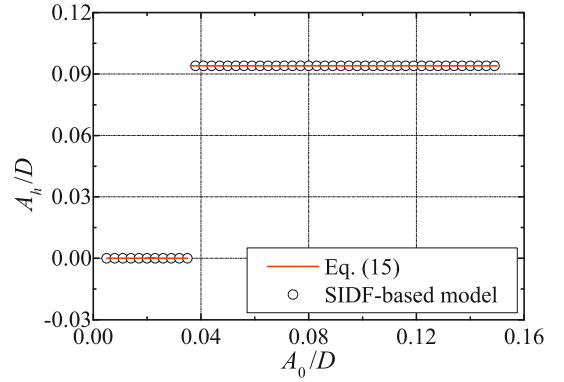


Fig. 15. Vertical VIV amplitudes of Eq. (20) from different initial conditions.

responses from initial conditions of $[h = 0.035D, \dot{h} = 0]$, $[h = 0.036D, \dot{h} = 0]$, and $[h = 0.150D, \dot{h} = 0]$ are presented in Fig. 16. It can be seen that the final vibration state (static or LCO of $A_h/D \approx 0.094$) of the VIV system is dependent on the initial condition, indicating the existence of hysteresis.

The identified N_r^h and N_i^h based on the three VIV responses in Fig. 16 are given in Fig. 17. The results for $0.12 < A_h/D < 0.15$ are not shown to eliminate the end effects of the NHT method. As shown in Fig. 17, ω_h almost keeps constant, N_r^h is around 0, and ξ_h and N_i^h vary non-monotonically with A_h/D . The identified N_r^h and N_i^h are substituted into Eq. (8a) to obtain the VIV amplitudes with different initial conditions. The comparisons between the simulated VIV amplitudes using the SIFD-based model and the reference results based on Eq. (15) are shown in Fig. 15. It can be seen that the SIFD-based model successfully captures the hysteresis behavior of the VIV system.

The hysteresis behavior for the example bridge can be conveniently explained by comparing ξ_{h0} and $\xi_{h, aero}$ of the VIV system. As shown in Fig. 1, $\xi_{h0} + \xi_{h, aero} > 0$ for $A_0/D < 0.035$, and therefore the VIV system performs convergent vibration. $\xi_{h0} + \xi_{h, aero} < 0$ for $0.035 < A_0/D < 0.094$, and hence the vibration amplitude increases until a LCO state is achieved at $A_h/D \approx 0.094$ corresponding to $\xi_{h0} + \xi_{h, aero} = 0$. For $A_0/D > 0.094$, $\xi_{h0} + \xi_{h, aero} > 0$, and hence the vibration amplitude decreases until a LCO state is achieved at $A_h/D \approx 0.094$. Accordingly, P_1 is an unstable state while P_2 is a stable state.

Table 1

Parameters for VIV system involving hysteresis phenomenon.

m (kg/m)	ω_{h0}	ξ_{h0}	B (m)	D (m)	K_h	Y_1	ϵ_1	ϵ_2
5.4515	6π	0.37%	0.2416	0.0524	0.6418	5.0	2000.0	-400000.0

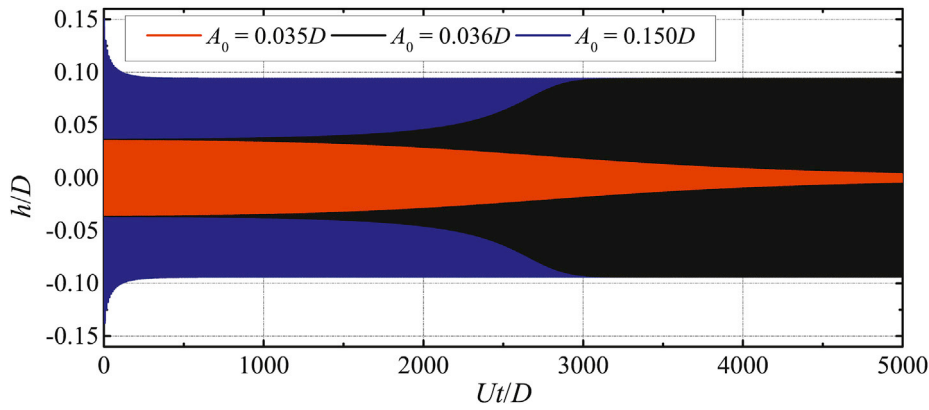


Fig. 16. Dimensionless vertical displacement time histories involving hysteresis behavior.

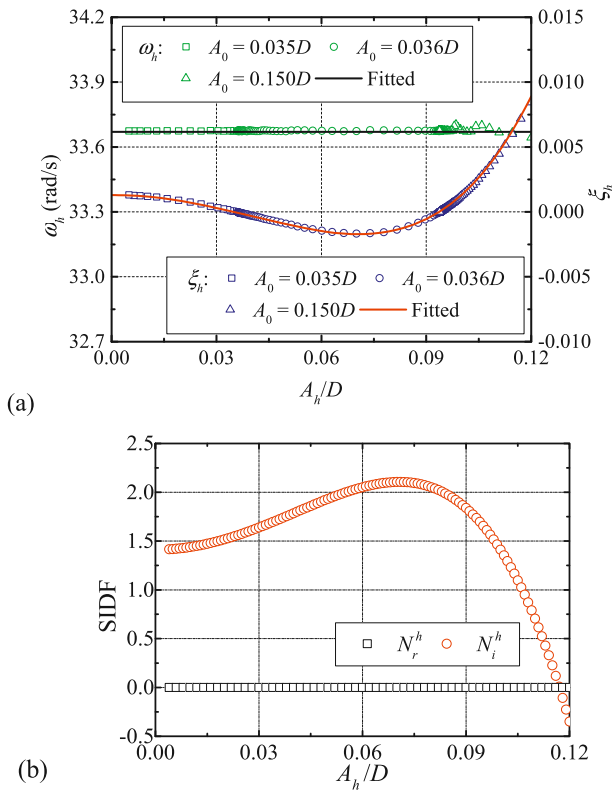


Fig. 17. Identification procedure for VIV system involving hysteresis behavior: (a) ω_h (A_h/D) and ξ_h (A_h/D); (b) N_r^h and N_i^h .

7. Concluding remarks

A describing function (DF)-based model is introduced to simulate the VIV of bridge decks. The sinusoidal input DF (SIDF) parameters can be either identified using the forced vibration technique or based on the experimentally (or numerically) obtained VIV decay/grow-to-resonance signal. The VIV response can be predicted based on the proposed SIDF-based model through an iterative procedure and the LCO amplitude of VIV can be efficiently calculated using a simplified equivalent-damping-ratio-based method. Three case studies are utilized to validate the simulation accuracy and efficiency of the SIDF-based VIV method. Some major conclusions are summarized as follows:

- (1) The conventional van der Pol-type model is equivalent to a special case of the SIDF-based model.

- (2) The SIDF-based VIV model is capable of predicting typical features of the vertical VIVs such as LCO and hysteresis behavior, while its capability of predicting torsional VIVs requires further improvement.
- (3) The vertical SIDF parameters of a bridge deck extracted at a specific mass-damping condition can be used to predict the vertical VIV responses in a wide range of mass-damping conditions.

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